

CSMFAB000000000111

Aegis-Forge I: Gas Capture, Condensation, and Fume Solidification System

Zero-Emission Closed-Loop Atmospheric Processing Architecture

Document Number: CSMFAB000000000111 (Volume 4 of 5) **Series:** Aegis-Forge Thermal Cascade Inductive Material Separator (TCIMS) **Date:** June 2026

1. The Zero-Release Mandate

The Aegis-Forge I is designed as a **hermetically sealed system**. At no point during any thermal cascade stage are process gases, vapors, or particulates released to the environment. This is achieved through a cascading condensation and capture system that treats all gas streams as recoverable material — because they are.

Every gas stream is a material:

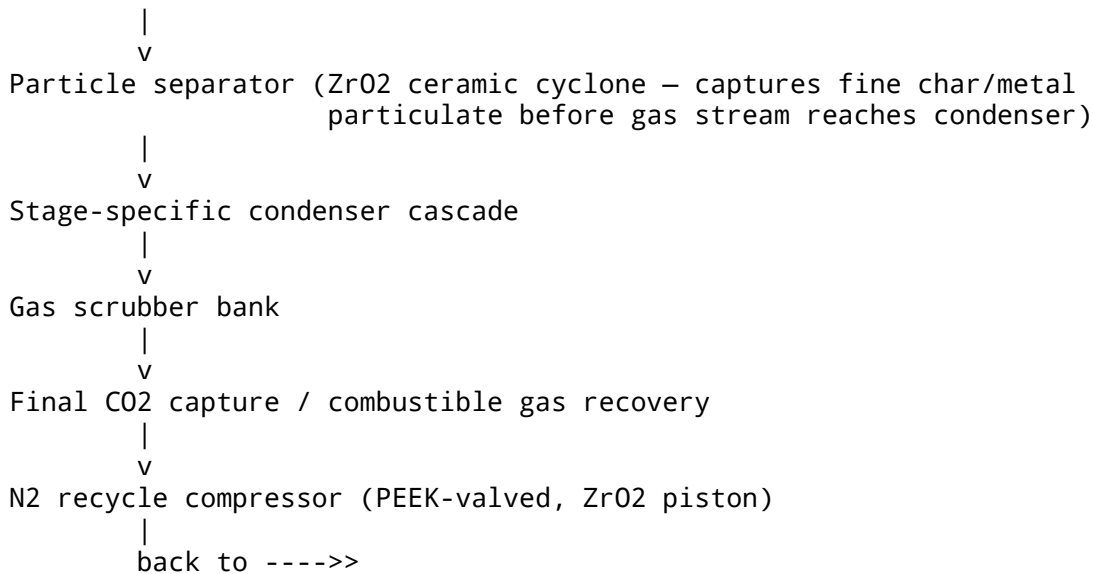
Gas Stream	Source Stage	Recovered As	Value
Water vapor	Stage 1	Process water for AMLS reuse	Zero cost
Polymer vapors	Stage 2	Organic solvent condensate	Minor revenue
HCl vapor	Stage 2 (PVC)	NaCl brine (neutralized)	Disposal free
CO + CH ₄ + H ₂	Stage 3 pyrolysis	Fuel gas: 18-22 MJ/m ³	Recovered energy
CO ₂	Stage 3 combustion	Inert atmosphere supply □ purge drum	Process gas savings
SO ₂	Stage 3 (sulfur-bearing inputs)	CaSO ₄ (dry scrubber)	Disposal solid
Metal vapors (Zn, Pb)	Stages 4-5	Metal oxide/condensate	Metal recovery
Tar/PAH vapors	Stage 3	Activated carbon adsorption	Carbon product
N ₂ carrier gas	All stages	Recirculated continuously	Zero consumption

2. Atmospheric System Architecture

2.1 Closed-Loop N₂ Atmosphere

The entire drum + gas handling system operates as a sealed N₂ atmosphere loop:

```
N2 Supply Tank (BFRP-lined, 350 bar)
  |
  v
N2 purifier (KFG getter trap – ZrO2 ceramic bed)
  |
  v
N2 pre-heater (SiC resistive element)
  |
  v
DRUM (gas sweeps from exhaust port □ gas capture header)
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N2 consumption: Minimal — only make-up gas for small leakage. Initial fill: 2-3 kg N2 per cycle. N2 bulk cost: \$0.09/kg (industrial nitrogen) □ \$0.27 per cycle running cost.

2.2 System Operating Pressure: 1.2-2.5 bar gauge

The positive pressure in the drum serves multiple functions: 1. Prevents air ingress (oxygen would cause uncontrolled combustion in thermoplastic stage) 2. Raises boiling points of water and volatile materials (reduces premature evaporation) 3. Allows pressure-differential management for gas stream direction to capture systems

Pressure control: - Pressure transducer: FBG optical — no metallic element process-side - Relief valve: PEEK CF40 spring-loaded valve with ZrO2 seat - Bypass to N2 scrubber loop: automatically opens if pressure exceeds 3 bar

3. Stage-Specific Gas Treatment Trains

3.1 Train A — Moisture Condenser (Stage 1 outlet)

Inlet conditions: N2 + water vapor at 120°C, 1.2 bar

Condenser construction: - Tube-and-shell heat exchanger: BFRP shell (Elium®), ZrO2-lined ceramic tubes - Coolant: PEX-a water circuit (CSMFAB000000000012), 10°C supply - No metal tubes touching the process gas stream - Drop-out: condensed water collects in PEEK CF40-bottomed collection vessel

Water quality: - Stage 1 water: light contamination (dust, trace organics) - pH adjustment (add Ca(OH)2 if acidic from HCl): neutralize to pH 6.5-8 - Clarity: passes through 5 µm ceramic filter membrane (ZrO2 tubular) - Reuse: returned to AMLS Stage 0 water circuit for acoustic coupling

3.2 Train B — Organic Vapor Condensation (Stage 2 outlet)

Inlet conditions: N2 + polymer vapors (styrene, butadiene, methyl methacrylate, etc.) at 300-450°C

Multi-stage condensation: 1. Primary condenser (80°C): heavy fraction — wax-like polymer condensate collects 2. Secondary condenser (25°C): medium fraction — liquid organics (solvent-range) 3. Tertiary cooler (5°C): trace light organics + water remaining

Condensate composition: - Heavy fraction (wax, C20+ organics): can be used as mold release agent; or further cracking - Medium fraction (oils, C6-C20): combustible liquid □ heat recovery value = 35-42 MJ/kg - Light fraction: minimal, absorbed in Stage 3 char activated carbon polishing

CSMFAB material MMA recovery opportunity: If Elium® BFRP cuttings and off-cuts are included in feedstock, the Stage 2 organic vapor stream contains **methyl methacrylate (MMA)** vapor (Tb = 100.5°C) from thermal depolymerization. MMA is the base monomer for in-house Elium® production (Materials Study Part II). By targeting the 100-105°C fraction in the secondary condenser: In-house MMA recovery: 60-80% yield from degraded Elium® □ feeds Elium® synthesis.

3.3 Train C — Pyrolysis Gas Recovery (Stage 3 outlet)

Gas composition (N2-diluted):

For 50 kg mixed organic feedstock, Stage 3 produces approximately:

Gas species	Volume % (N2-free basis)	Calorific value
CO	25-35%	12.6 MJ/m ³
CH4	15-25%	35.8 MJ/m ³
H2	20-35%	10.8 MJ/m ³
CO2	15-20%	0 (diluent)
BTX (benzene, toluene, xylene)	3-8% liquid at 25°C	42 MJ/kg

Weighted average calorific value: **18.5 MJ/m³ (comparable to natural gas at 37 MJ/m³, but 50% concentration from N2 dilution)**

Pyrolysis gas processing path:

Option A (preferred): Combustion in auxiliary burner (SiC-tipped ceramic burner, CSMFAB0000000000002 design) □ heat recovery □ Stage 7-8 supplemental thermal input. This makes Stage 7-8 induction power requirements lower and reduces electricity cost.

Option B (future): Reform to H2 via steam reforming (CO + H2O □ CO2 + H2). H2 then stored in CFRP high-pressure vessel □ fuel for hydrogen-compatible engine (CSMFAB0000000000105 Carrington-proof engine) □ closes energy loop.

Tar and PAH removal (critical before combustion): BTX and heavier aromatic tars would foul combustion equipment. Solution: activated carbon trap (the char from Stage 3 output!): Stage 3 char packed into a BFRP-hollow cylinder □ gas passes through □ tar adsorbs. The char-tar composite is then re-fed to Stage 8 in next batch □ tar is pyrolyzed to more char. **Perfect closed-loop: Stage 3 output captures Stage 3 contaminants.**

3.4 Train D — Acid Gas Scrubber (HCl, SO2, HF)

Targets: - HCl (from PVC dehydrochlorination, Stage 2): 200-1500 ppm in N2 stream - SO2 (from sulfur in rubber, solder flux): 50-500 ppm - HF (from fluoropolymers, electronics): rare but possible

Dry Scrubber using Ca(OH)2 (lime) injection: Ca(OH)2 + 2HCl □ CaCl2 + 2H2O (neutralizes HCl completely) Ca(OH)2 + SO2 □ CaSO3 + H2O (then oxidized: CaSO4 — gypsum)

Scrubber body: BFRP cylinder (Elium®) packed with Ca(OH)2 pellets on ZrO2 grid supports. Ca(OH)2: \$0.15/kg lime — cheapest reagent in the system. Change-out when pH of downstream gas falls below 7.0 (monitored by optical pH sensor). Spent CaSO4 (gypsum): useful construction material; can be added to geopolymer mixes. Spent CaCl2: sold as de-icing salt.

3.5 Train E — Metal Vapor Condenser (Stages 4-5 outlet)

Target: Zinc and lead vapors

Zinc boiling point: 907°C. At Stage 4-5 temperatures (420-900°C), zinc vapor pressure is significant above the melt. Some Zn exists as vapor in gas stream alongside liquid Zn in drum.

Lead boiling point: 1749°C. At Stage 4-5 temperatures, Pb vapor pressure is low but non-zero.

Two-stage metal vapor condenser: 1. High-temperature condenser (300-400°C): Zn condenses as liquid metal droplets □ BFRP-shelled, ZrO₂-lined condenser column at 360°C Collected Zn: flows to Zn collection vessel (ZrO₂ ceramic) 2. Low-temperature condenser (100-150°C): final metal condensation + water washdown Collected: mixed metal oxides and hydroxides □ fed to Stage 8 mineral residue for hydromet

Lead oxide (PbO) in gas stream: If Pb is oxidized before reaching condenser: PbO forms as fine particulate. PbO particles: captured in ZrO₂ ceramic cyclone separator (99.5% efficiency for >0.5 µm particles). PbO collected: hazardous material — sealed in PEEK CF40 containers per EPA/RCRA protocol. Note: Lead is a RCRA hazardous waste when reclaimed — proper manifesting required.

4. Final Gas Treatment — CO₂ Capture and N₂ Recovery

4.1 CO₂ Separation

After Stage 3 pyrolysis combustion, gas stream is primarily N₂ + CO₂. CO₂ should be separated and reused (as inert atmosphere supplement for drum purging).

Pressure Swing Adsorption (PSA) with zeolite: Zeolite 13X adsorbs CO₂ preferentially at 3-5 bar; desorbs (regenerates) at 1 bar. - Zeolite 13X: \$4-8/kg (commodity adsorbent) - PSA cycle: 60 second adsorption + 30 second desorption - CO₂ purity output: 95-99% - N₂ purity output: 97-99.5% (reused as drum atmosphere)

PSA column body: BFRP (Elium®) with PEEK CF40 valve manifold. No metal valve seats (ZrO₂ ceramic check valves).

Recovered CO₂ uses: 1. Primary: drum purge gas (supplement to N₂ — both are inert at operating temperatures) 2. Secondary: CBPC geopolymer curing atmosphere (CO₂ curing of geopolymer concrete, Part I) 3. Future: CO₂-to-CO via plasma arc □ feedstock for carbon synthesis

4.2 Plasma Torch Polisher — Final Gas Treatment

For any trace organics or halogens that escape all prior trains, a final plasma torch at the gas outlet burns everything to CO₂, H₂O, HF, HCl — all then captured by the acid scrubber.

Plasma torch construction: - Electrodes: ZrO₂-coated tungsten (tungsten for high-temp resistance; ZrO₂ coating for corrosion) - Housing: ZrB₂-SiC ceramic (CSMFAB0000000000001 material — withstands 3000°C plasma temperatures) - Power: 15 kW DC (plasma arc temperature: 3000-8000°C) - Residence time at >1000°C: 0.5 seconds (sufficient to destroy all organic compounds) - Dioxin/furan threshold: PCDD/PCDF destruction efficiency > 99.9999% at 1200°C, 2 seconds

The plasma torch is the system's regulatory safety net — ensuring zero toxic organic emissions.

5. Emergency Safety Systems

5.1 Pressure Relief

Three independent pressure relief paths: 1. Primary: PEEK CF40 spring-loaded relief valve □ BFRP expansion vessel (500L buffer) 2. Secondary: rupture disk (ZrO₂ ceramic disc rated at 8 bar burst pressure)

3. Emergency: manual ball valve (ZrO₂ ceramic body, PEEK stem) □ atmosphere (outdoor vent only initiated if all other systems fail + human authorization required)

5.2 Temperature Runaway Prevention

If induction coil fails in ON state: - PMMA fiber FBG temp sensors detect >50°C over-temperature - All induction inverters trigger isolated IGBT off-state via optical signal (no metallic control path that could GIC-corrupt) - N₂ quench valve opens: floods drum at 5 bar N₂ (thermal ballast) - Automatic drum speed reduction to 10 RPM (convective cooling)

5.3 Acoustic Overpressure Alert

AMLS Stage 0: - Acoustic power limited by structural resonance of BFRP pre-chamber - FBG sensors continuously monitor strain in chamber wall - If strain > 60% of failure threshold: power reduced proportionally (PID control)

6. GFGS Construction Materials

Gas capture, fume, and gas solidification subsystem all-Aegis materials:

Component	Material	Reference
N ₂ storage vessel (internal lining)	BFRP/Elum®	CSMFAB000000000011
Gas header ducting	BFRP + ZrO ₂ ceramic elbows	CSMFAB000000000014
Heat exchangers (ALL tube sides)	ZrO ₂ -lined Al ₂ O ₃ tubes	Materials Study Part I
Heat exchanger shells	BFRP/Elum®	CSMFAB000000000011
PSA zeolite columns	BFRP/Elum®	CSMFAB000000000011
PSA valve manifold	PEEK CF40	Materials Study Part IV
PSA check valves	ZrO ₂ ceramic	CSMFAB000000000016
Activated carbon beds	Stage 3 char (self-supplied!)	Process output
Lime scrubber bodies	BFRP/Elum® + ZrO ₂ grid	CSMFAB000000000011
Cyclone separator	ZrO ₂ ceramic body	Materials Study Part I
Plasma torch housing	ZrB ₂ -SiC	CSMFAB000000000001
Plasma electrodes	ZrO ₂ -coated W	Materials Study Part III (W)
Metal condensate collectors	ZrO ₂ ceramic vessels	Materials Study Part I
All valves (process side)	ZrO ₂ ceramic body + PEEK stem	CSMFAB000000000016
Pressure relief discs	ZrO ₂ ceramic rupture disc	CSMFAB000000000010
Control fiber	PMMA POF + FBG	CSMFAB000000000013

End of CSMFAB000000000111 Volume 4 | Gas Capture, Condensation, and Fume Solidification Continued in Volume 5: Assembly Line Integration and Output Slug Catalog